

flow network $G=(V,E,c)$
 $s,t \in V$

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Lemma: Let $f: E \rightarrow \mathbb{R}$ be any feasible
~~flow~~ (s,t) -flow and let (S,T)
be any (s,t) -cut. Then:

① $|f| \leq \|S,T\|$

② $|f| = \|S,T\| \iff f$ saturates every
edge from **S** to **T** +
"avoids" every edge from
T to S.

Proof: Let f be a feasible flow
and (S,T) be a cut.

Then,

1: $|f| = \partial f(s) \leftarrow$ by definition

2: $= \sum_{v \in S} \partial f(v) \leftarrow \begin{matrix} t \notin S \\ \partial f(v) = 0 \forall v \in V \setminus \{s,t\} \end{matrix}$

3: $= \sum_{v \in S} \sum_w f(v \rightarrow w) - \sum_{v \in S} \sum_u f(u \rightarrow v)$ by def'n of ∂

4: $= \sum_{v \in S} \sum_{w \in S} f(v \rightarrow w) - \sum_{v \in S} \sum_{u \notin S} f(u \rightarrow v)$ see next page.

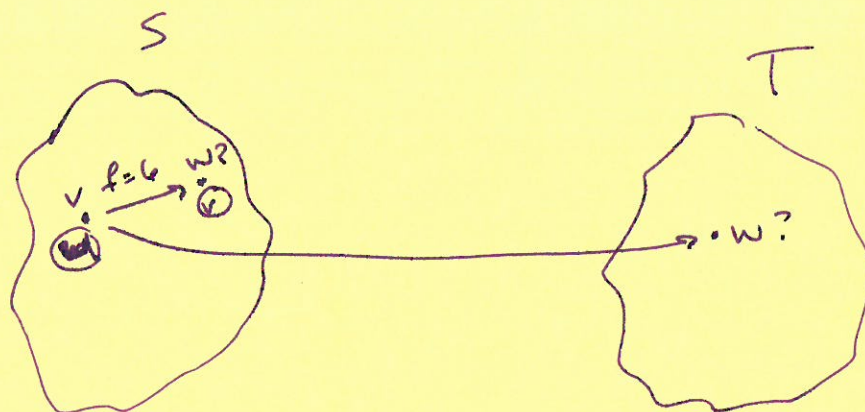
5: $= \sum_{v \in S} \sum_{w \in T} f(v \rightarrow w) - \sum_{v \in S} \sum_{u \in T} f(u \rightarrow v)$ note:
 $u, w \in V$

6: $\leq \sum_{v \in S} \sum_{w \in T} f(v \rightarrow w)$ b/c f is feasible

7: $\leq \sum_{v \in S} \sum_{w \in T} c(v \rightarrow w)$ by definition of cut.

8: $= \|S,T\|$

□ ①



Why $L_3 = L_4$:

each edge in S appears 1x in pos sum

$$\sum_{S \in \mathcal{S}} \sum_{w \in V} f(v \rightarrow w)$$

and 1x in neg sum

$$- \sum_{S \in \mathcal{S}'} \sum_{u \in V} f(u \rightarrow v)$$

Line 3:

$$\sum_{v \in \mathcal{S}} \sum_{w \in V} f(v \rightarrow w) - \sum_{v \in \mathcal{S}'} \sum_{u \in V} f(u \rightarrow v)$$

$$= \sum_{v \in \mathcal{S}'} \left(\sum_{w \in \mathcal{S}} f(v \rightarrow w) + \sum_{w \in T} f(v \rightarrow w) \right) - \sum_{v \in \mathcal{S}} \left(\sum_{u \in \mathcal{S}} f(u \rightarrow v) + \sum_{u \in T} f(u \rightarrow v) \right)$$

by def'n of partition

$$= A + B - A - C$$

$$= \sum_{v \in \mathcal{S}} \sum_{w \in T} f(v \rightarrow w) - \sum_{v \in \mathcal{S}'} \sum_{u \in T} f(u \rightarrow v)$$

by cancellation

↑ Line 4

Theorem: The value of a maximum flow in a flow graph is equal to the cost of a minimum cut in that graph.

"max flow = min cut"

Proof: (The crux)

Residual graphs, augmenting paths
+ the fact that

"If there does not exist an
(s,t)-path in the residual
graph using ~~non~~ positive edges only,
then the flow is maximized."